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Coupling structure of rear end module and quarter rear member

Abstract

Provided is a coupling structure of a rear end module and a quarter rear member. The coupling structure includes a quarter rear member located on a rear portion of a side surface of a vehicle; a rear floor panel in which rear side members arranged in a longitudinal direction of the vehicle are bonded to both side surfaces of a bottom surface; a connection member having an upper portion bonded to one side of the quarter rear member and a lower end bonded to the rear floor panel; and a rear end module formed by injection to have an annular shape and coupled to a rear end of the connection member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims under 35 U.S.C. § 119(a) the benefit of Korean Patent Application No. 10-2022-0028636, filed on Mar. 7, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) Embodiments of the present disclosure relate to a coupling structure of a rear end module and a

quarter rear member, in which the rear end module is coupled to a quarter rear member located on a rear portion of a side surface of a vehicle body and a member adjacent thereto on a rear end of a vehicle.

BACKGROUND

(3) Sport utility vehicle (SUV), recreational vehicle (RV), etc. have a tailgate installed on a rear end of the vehicle, so that the rear end of the vehicle is opened and closed by opening and closing the tailgate.

(4) The rear end of the vehicle is finished with a rear end part **120**, and when the tailgate is closed, a circumference of the tailgate is in contact with a circumference of the rear end part **120**. A lower corner part **121** of the rear end part is configured so that the rear end part is connected to a rear floor panel **116**, a wheelhouse **115**, and a quarter rear member **117**, and reinforcing members **131**, **132** for reinforcing rigidity are applied to the connection portions thereof.

(5) However, according to the related art, the reinforcing members **131**, **132** are not connected to each other, and the wheelhouse **115** and the quarter rear member **117** are not connected, so that a load path cannot be formed. As shown in FIG. 3, after the load input from the rear floor panel **116** is transmitted to the reinforcing member **131** applied between the rear end part and the rear floor panel **116**, and then the load is transmitted to the reinforcing member **132** applied between the rear end part and the quarter rear member **117**, but the load is not sufficiently transmitted.

(6) As described above, since the load path is not formed in the lower corner part **121** of the rear end part, the load is not sufficiently distributed and thus the load is partially concentrated, which is a factor causing a reduction in rigidity.

(7) The contents described in Description of Related Art are to help the understanding of the background of the present disclosure, and may include what is not previously known to those skilled in the art to which the present disclosure pertains.

SUMMARY OF THE DISCLOSURE

(8) Embodiments of the present disclosure relate to a coupling structure of a rear end module and a quarter rear member, which allows the rear end module coupled to a rear end of a vehicle to be formed in an integrated annular structure and allows the quarter rear member to be connected to a lower end of the rear end module while maintaining a closed cross section, thereby improving rigidity.

(9) A coupling structure of a rear end module and a quarter rear member according to embodiments of the present disclosure includes a quarter rear member located on a rear portion of a side surface of a vehicle; a rear floor panel in which rear side members arranged in a longitudinal direction of the vehicle are bonded to both side surfaces of a bottom surface; a connection member having an upper portion bonded to one side of the quarter rear member and a lower end bonded to the rear floor panel; and a rear end module formed by injection to have an annular shape and coupled to a rear end of the connection member.

(10) In some embodiments, the connection member has middle and lower portions, other than the upper portion, spaced apart from the quarter rear member, and the lower end bonded to the rear floor panel, so that a closed cross section is formed between the quarter rear member, the connection member, and the rear floor panel.

(11) In some embodiments, the connection member is formed to be bent toward an inside of the vehicle on the rear end of the side surface of the vehicle along a circumference of the rear floor panel.

(12) In some embodiments, a reinforcing member is provided on one side of a front surface of the rear end module in order to reinforce rigidity between the rear end module and the connection member.

(13) In some embodiments, the rear end module is injected by an over-mold method and coupled to the reinforcing member, so that a closed cross section is formed on the rear end module.

(14) In some embodiments, a fastening bolt is fastened to the rear end module by penetrating the

upper portion of the connection member and the reinforcing member.

(15) In some embodiments, a back panel is interposed between the connection member and the reinforcing member.

(16) In some embodiments, a rear end of the rear side member is bonded to the reinforcing member and the rear end module by an adhesive.

(17) In some embodiments, a side end of the rear floor panel is bent downward and bonded to a lower portion of the quarter rear member.

(18) In some embodiments, in a portion of the quarter rear member where the rear floor panel is bonded, a wheel housing is bonded to a side opposite to the rear floor panel.

(19) In some embodiments, ribs are radially formed inside the rear end module.

(20) According to the coupling structure of the rear end module and the quarter rear member according to embodiments of present disclosure having the above configuration, it possible to connect the wheel housing and the quarter rear member through the connection member to form the closed cross section, thereby improving rigidity, and to connect the closed cross section to the rear end module to form the load path, thereby distributing the load.

(21) In some embodiments, the connection member is bent may be coupled to the reinforcing member.

(22) In some embodiments, the reinforcing member is made of a metallic material.

(23) In some embodiments, the rear end module is injected with a synthetic resin.

(24) In some embodiments, the rear side member is bonded to a bottom surface of the rear floor panel.

(25) As described above, the structure in which the load is distributed is configured, thereby increasing connectivity on the corner part of the rear end module and improving skeletal rigidity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front diagram of a rear end part according to the related art.

(2) FIG. 2 is a detailed diagram of portion I of FIG. 1.

(3) FIG. 3 is a schematic diagram showing a load path on the rear end part according to the related art.

(4) FIG. 4 is an exploded perspective diagram showing a state where a rear end module is coupled to the rear end part according to a coupling structure of the rear end module and a quarter rear member according to an embodiment of the present disclosure.

(5) FIG. 5 is a cross-sectional diagram taken along line A-A of FIG. 4.

(6) FIG. 6 is a detailed diagram showing the rear end module with respect to portion II of FIG. 4.

(7) FIG. 7 is a detailed diagram showing a portion where the rear end module is coupled with respect to the portion II of FIG. 4.

(8) FIG. 8 is a side diagram of the coupling structure of the rear end module and the quarter rear member according to an embodiment of the present disclosure.

(9) FIG. 9 is an enlarged bottom diagram of a main part of the coupling structure of the rear end module and the quarter rear member according to an embodiment of the present disclosure.

(10) FIG. 10 is a cross-sectional diagram taken along line B-B of FIG. 4.

(11) FIG. 11 is a cross-sectional diagram taken along line C-C of FIGS. 8 and 9.

DESCRIPTION OF SPECIFIC EMBODIMENTS

(12) Hereinafter, a coupling structure of a rear end module and a quarter rear member according to embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

(13) A coupling structure of a rear end module and a quarter rear member according to

embodiments of the present disclosure may include a quarter rear member **17** located on a rear portion of a side surface of a vehicle, a rear floor panel **16** in which a rear side member **14** arranged in a longitudinal direction of the vehicle is bonded to both sides of a bottom surface, a connection member **18** having an upper portion bonded to one side of the quarter rear member **17** and a lower end bonded to the rear floor panel **16**, and a rear end module **20** formed by injection to have an annular shape and coupled to a rear end of the connection member **18**.

(14) The rear end module **20** formed by injection to have an annular shape may be coupled to the rear end part of the vehicle.

(15) The rear end module **20** may be integrally formed by using a synthetic resin as a material. The rear end module **20** may be integrally formed in an annular shape, and is coupled to the rear end part of the vehicle. When a tailgate is closed, the rear end module **20** may become a portion in contact with the tailgate.

(16) Ribs **20a** may be formed inside the rear end module **20** to improve rigidity. The ribs **20a** each may be radially formed, and a fastening bolt **25** to be described later may be fastened to a portion where the ribs **20a** intersect.

(17) The rear end part fastened to the rear end module **20** may become a rear end of a rear structure of a vehicle body composed of a roof panel, a roof side member **12**, a middle member **13**, the rear floor panel **16**, the rear side member **14**, the quarter rear member **17**, and a wheelhouse **15**.

(18) In particular, according to embodiments of the present disclosure, when the rear end module **20** is coupled to a lower end of the rear end part, that is, a rear end of each of the rear floor panel **16**, the rear side member **14**, the quarter rear member **17**, and the wheelhouse **15**, a load path is efficiently formed, thereby improving the rigidity of the vehicle body. A load acting in the longitudinal direction of the vehicle may be transmitted from a C-pillar member **11** to the rear end module **20** through the roof side member **12** and the middle member **13**, or transmitted to the rear end module **20** through the wheelhouse **15** and the quarter rear member **17** (see the arrow in FIG. 4).

(19) The rear end module **20** may be connected to a rear end of the quarter rear member **17** located on a rear portion of a side surface of the vehicle and a rear end of the rear floor panel **16** forming a rear floor of the vehicle. At this time, the quarter rear member **17** and the rear floor panel **16** may be connected to the rear end module **20** through the connection member **18**.

(20) Meanwhile, a reinforcing member **21** for reinforcing rigidity may be provided on a portion of the connection member **18** connected to the rear end module **20**.

(21) The reinforcing member **21** may be made of a metallic material such as a steel, unlike the rear end module **20** injected with a synthetic resin. When the rear end module **20** is injected, the rear end module **20** may be injected by an over-mold method so that the reinforcing member **21** is integrated with the rear end module **20**. In a state where the reinforcing member **21** is disposed, the rear end module **20** may be injected by the over-mold method by injecting a synthetic resin, so that the reinforcing member **21** and the rear end module **20** are integrated with each other. In addition, in the portion of the rear end module **20** where the reinforcing member **21** is located, a closed cross section S may be formed between the rear end module **20** and the reinforcing member **21** by the over-mold method. The closed cross section may be connected to the closed cross section S formed by the connection member **18**, so that rigidity is reinforced by the closed cross section S. In other words, the connecting member **18** may form the closed cross section S between the connection member **18**, the quarter rear member **17**, and a back panel **19** as the connection member **18** is connected to the quarter rear member **17** and the back panel **19** (see FIGS. 5 and 10), and the thus formed closed cross section S may be located on the same line as that of the closed cross section S formed by the over-mold method in the rear end module **20** to form the load path.

(22) An upper end of the connection member **18** may be bonded to the quarter rear member **17**. Middle and lower portions of the connection member **18** may be spaced apart from the quarter rear member **17**. The lower end of the connection member **18** may be bonded to the rear floor panel **16**.

The rear floor panel **16** may be bonded to the lower portion of the quarter rear member **17**. Accordingly, the closed cross section **S** may be formed by the quarter rear member **17**, the connection member **18**, and the rear floor panel **16** (see FIG. 5).

(23) The connection member **18** may be bent toward the inside of the vehicle from a rear end of the side surface of the vehicle along a circumference of the rear floor panel **16**. A portion where the connection member **18** is bent may be coupled to the reinforcing member **21**. At this time, the back panel **19** may be interposed between the connection member **18** and the reinforcing member **21**, and as a gap is formed between the connection member **18** and the back panel **19**, the closed cross section **S** is formed therebetween (see FIG. 11). For the closed cross section formed between the connection member **18** and the back panel **19**, the closed cross section **S** formed by the quarter rear member **17**, the connection member **18**, and the rear floor panel **16** may form the load path together with the closed cross section **S** formed by the over-mold method in the rear end module **20** (see the arrows located on the tops in FIGS. 8 and 9).

(24) The connection member **18** may be fastened to the reinforcing member **21** and the rear end module **20** through a fastening bolt **25**. The fastening bolt **25** may be fastened to the rear end module **20** by penetrating the upper portion of the connection member **18** and the reinforcing member **21**. Since the connection member **18** is bonded to the quarter rear member **17** and the rear floor panel **16**, the quarter rear member **17** and the rear floor panel **16** may have the same effect as that of being fastened to the reinforcing member **21** and the rear end module **20**.

(25) The rear side member **14** may be bonded to a bottom surface of the rear floor panel **16**. The rear side member **14** may have a structure in which the closed cross section is formed therein, and the rear side member **14** may be connected to the lower portion of the reinforcing member **21** to form another load path (see the arrows located at the bottoms in FIGS. 8 and 9). An adhesive **26** may be applied to the rear end of the rear side member **14** and is bonded to the lower portion of the reinforcing member **21** (e.g., a portion where a through hole is formed in FIG. 6) by the adhesive **26**.

Claims

1. A coupling structure of a rear end module and a quarter rear member comprising: a quarter rear member configured to be located on a rear portion of a side surface of a vehicle; a rear floor panel; a connection member having an upper portion bonded to one side of the quarter rear member and a lower end bonded to the rear floor panel; a rear end module formed by injection to have an annular shape and coupled to a rear end of the connection member; a reinforcing member provided on one side of a front surface of the rear end module in order to reinforce rigidity between the rear end module and the connection member; a back panel interposed between the connection member and the reinforcing member; wherein a first closed cross section is formed between the connection member, the quarter rear member, and the back panel; wherein a second closed cross section is formed between the rear end module and the reinforcing member; and wherein the first closed cross section connects the second closed cross section to form a load path.
2. The coupling structure of claim 1, wherein the connection member has middle and lower portions, other than the upper portion, spaced apart from the quarter rear member, and the lower end bonded to the rear floor panel, so that a third closed cross section is formed between the quarter rear member, the connection member, and the rear floor panel.
3. The coupling structure of claim 1, wherein the connection member is bent toward an inside of the vehicle on the rear end of the side surface of the vehicle along a circumference of the rear floor panel.
4. The coupling structure of claim 1, wherein the connection member is bent may be coupled to the reinforcing member.
5. The coupling structure of claim 1, wherein the reinforcing member is made of a metallic

material.

6. The coupling structure of claim 1, wherein the rear end module is injected with a synthetic resin.
 7. The coupling structure of claim 1, wherein the rear end module is injected by an over-mold method and coupled to the reinforcing member, so that the closed cross section is formed on the rear end module.
 8. The coupling structure of claim 1, wherein a fastening bolt is provided to fasten to the rear end module by penetrating the upper portion of the connection member and the reinforcing member.
 9. The coupling structure of claim 1, wherein a rear end of a rear side member is bonded to the reinforcing member and the rear end module by an adhesive.
 10. The coupling structure of claim 1, wherein a side end of the rear floor panel is bent downward and bonded to a lower portion of the quarter rear member.
 11. The coupling structure of claim 10, wherein in a portion of the quarter rear member where the rear floor panel is bonded, a wheel housing is bonded to a side opposite to the rear floor panel.
 12. The coupling structure of claim 1, wherein ribs are radially formed inside the rear end module.
 13. The coupling structure of claim 1, wherein a rear side member is bonded to a bottom surface of the rear floor panel.
 14. A vehicle comprising the coupling structure of claim 1.
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